

Chapter 131 — Management of Muscle-Invasive and Metastatic Bladder Cancer

- **Burden:** in the United States in 2022, an estimated **81,000 new cases and 17,000 deaths**; bladder cancer is the **fourth commonest malignancy in males** (about 6% of all cancer diagnoses). Less common in females, yet **female sex is associated with worse survival**.
- **Stage at presentation:** most patients have non-muscle-invasive bladder cancer (NMIBC); **20–30% present with muscle-invasive bladder cancer (MIBC)** and **about 12% with node-positive or metastatic disease**.
- **Care is multidisciplinary** — urologist, medical oncologist and radiation oncologist — from diagnosis onwards.

NATURAL HISTORY

- **Primary versus secondary MIBC:** most MIBC is muscle-invasive at diagnosis (**primary**); about **20% progress from NMIBC** during initial management (**secondary**).
- **Five-year survival:** localised disease about 70% (a figure that includes NMIBC); **regional disease about 39%; distant disease about 8%**.
- **Secondary MIBC does worse.** Initial understaging contributes, but secondary MIBC also shows **lower pathological response to neoadjuvant chemotherapy (NAC)** — evidence of a **biological and possibly genomic difference** from primary MIBC.
- **Death after adequate local therapy is systemic:** most deaths occur **within 2 years** of treatment, and non-locoregional relapse reflects **micrometastatic disease already present at diagnosis**. Hence the central place of **neoadjuvant and adjuvant (multimodal) therapy**.

Risk Factors

- **Tobacco is the number one risk factor**, accounting for **nearly half of all bladder cancers**; the carcinogens are **aromatic amines and N-nitroso compounds**.
 - Smokers carry a **relative risk of about 3.5** versus non-smokers; risk rises with **amount and duration** smoked and **falls after cessation**.
 - **Screen every patient for smoking status and advise cessation at diagnosis**. The effect of cannabis, e-cigarettes and vaping is **unknown**.
- **Occupational exposure:** aluminium production, rubber production, dye work, coal-tar pitch, dry cleaning, hairdressing, printing and textile manufacture; carcinogens range from **2-naphthylamine** to **4,4'-methylene-bis compounds** and **tetrachloroethylene**. Occupation accounts for **about 7% of cases in males**; **no association has been found in females**.
- **Other factors** (detailed in Chapter 129): radiation, environmental exposures, **chronic cystitis and schistosomiasis (both squamous cell carcinoma)**, **cyclophosphamide and pioglitazone**, genetic and dietary factors.

Concept — Muscle-invasive bladder cancer is **a systemic disease at diagnosis in a large minority**: death after local therapy comes from micrometastases already present, which is why every curative pathway now wraps local treatment in systemic therapy.

HISTOLOGY

- **Urothelial carcinoma (UC)** accounts for **90%** of bladder tumours.

- **Radical cystectomy (RC) is standard for MIBC whatever the histology;** what varies by subtype is the **use and timing of NAC and/or radiation.**
- **Variant histology at RC** is more often **upstaged**, with more **extravesical disease, lymphovascular invasion (LVI) and nodal metastasis** than pure UC — so it must be **recognised early.**
- **American Urological Association (AUA): genitourinary pathologist review** when variant histology is suspected or muscle invasion is equivocal.

VARIANT HISTOLOGIES COMPARED

Histology	Frequency and setting	Associations and pitfalls	Treatment
Squamous cell carcinoma	5% in the West; commoner in the Middle East and Africa (endemic schistosomiasis), where its share is falling (effective anti-bilharzial drugs, less smoking)	Chronic cystitis, chronic catheterisation	Poor response to chemotherapy and radiation → up-front RC and lymphadenectomy
Adenocarcinoma	2%; arises from urothelium or a patent urachus; commonest tumour in bladder exstrophy	May be secondary — exclude metastasis from colorectal and breast primaries	Poor response to systemic therapy and radiation → RC and lymphadenectomy (urachal: partial cystectomy — see Partial Cystectomy)
Neuroendocrine (usually small cell; large cell behaves similarly)	Rare, aggressive; high stage or metastatic at presentation	Paraneoplastic: ectopic adrenocorticotrophic hormone (ACTH), hypercalcaemia, hypophosphataemia	Chemosensitive → NAC (etoposide + cisplatin), then RC or consolidative local therapy
Micropapillary	Resembles ovarian papillary serous carcinoma; high grade, high stage	Frequent LVI; NAC response data conflicting	Predominant micropapillary → RC
Mixed UC with variant	—	Quantify the variant component	Predominant UC → NAC may benefit; predominant variant → up-front RC

- **Other variants** — nested/microcystic, large nested, glandular, clear cell, microtubular, plasmacytoid, lymphoepithelioma-like, giant cell and sarcomatoid/carcinosarcoma — are **all high grade under the World Health Organization (WHO) 2022 classification.** Carcinoid tumours rarely arise in or invade the bladder.
- **Rare non-urothelial entities:** rhabdomyosarcoma, leiomyosarcoma, primary lymphoma.
- Evidence for most variants is scant, but most are aggressive: **consider early definitive RC with lymphadenectomy.**

Concept — The dividing line between variants is **chemosensitivity:** neuroendocrine and urothelial-predominant tumours earn neoadjuvant chemotherapy, whereas squamous, adenocarcinoma and predominantly micropapillary tumours go straight to cystectomy.

CLINICAL STAGING AND DIAGNOSIS

- **Clinical stage** estimates disease burden before RC; **pathological stage** comes from microscopy of the RC and pelvic lymphadenectomy specimens.
- **Clinical staging combines:** depth of invasion on transurethral resection (TUR) → bimanual examination under anaesthesia (EUA) → abdominopelvic cross-sectional imaging with and without contrast, with excretory phase for the upper tracts → chest imaging.

- **Clinical-pathological discordance is common:** in clinical T2 disease, pathological **upstaging occurs in about 40–54%** and downstaging in about 17%.
 - **Predictors of upstaging:** **LVI, variant histology, female sex;** LVI also predicts nodal disease at RC; **secondary MIBC is upstaged more often** than primary.
 - **Understaging increases progression and death from disease.**

Transurethral resection

- **TUR is the standard diagnostic procedure;** a genitourinary pathologist should report depth of invasion and histological subtype.
- **Document** the location, size, number and appearance of tumours; consider a **separate deep specimen from each tumour**, especially if invasive (TUR technique: Chapter 132).
- **Resect all macroscopic tumour when safe and feasible:** this lowers local burden, may **optimise NAC response**, and improves the chance of **successful bladder preservation with chemoradiotherapy**.
- **pT0 at RC confers a survival benefit in randomised NAC trials whether or not NAC was given** — suggesting a benefit from **aggressive TUR** itself.

The prostatic urethra and the female urethra

- **Inspect the prostatic urethra (males) and urethra (females) at TUR.**
- **Risk factors for prostatic urethral involvement:** tumour at the **trigone or bladder neck**, carcinoma in situ (CIS), high-volume multifocal disease.
- **Why it matters:** **prostatic stromal involvement** influences the need for **NAC**, the need for **urethrectomy**, and the **choice of urinary diversion**.
- **Prostatic urethral biopsy at TUR:** full loop of tissue from mid-prostate to the mid-to-distal verumontanum → at **5 and 7 o'clock adjacent to the verumontanum** (highest density of prostatic ducts; commonest site of CIS) → plus any abnormal-looking tissue. A **full-thickness** resection shows the pathologist the interface of urethral mucosa, prostatic ducts and stroma, permitting accurate staging.
- **Frozen section at cystoprostatectomy:**
 - **Negative TUR biopsy:** the European Association of Urology (EAU) suggests the intraoperative frozen margin may be omitted; however, **intraoperative frozen section has a higher negative predictive value**, so the authors **recommend it whenever orthotopic diversion is considered**.
 - **Positive TUR biopsy** → **intraoperative frozen urethral margin is required; if negative, orthotopic diversion can be considered**.
- **Females:** urethral involvement is rare; take a **proximal urethral margin whenever orthotopic diversion is considered**. Risk factors: involvement of the **bladder neck, bladder base and anterior vaginal wall**.

Bimanual examination under anaesthesia

- Perform **before and after resection**, with the **bladder drained and no catheter in place**; dominant hand suprapubic, one or two fingers of the other hand in the rectum (male) or vagina (female); assess **palpable disease and bladder mobility**.

CLINICAL T STAGE ON BIMANUAL EXAMINATION

Clinical stage	Finding
T2a	Nonpalpable
T2b	Induration without a three-dimensional mass
T3a	Three-dimensional mass, mobile
T4a	Invades adjacent structures (prostate, vagina, rectum)

Clinical stage	Finding
T4b	Fixed to the pelvic sidewall , immobile

- **Interpret with caution:** bimanual examination both overstages and — more often — **understages**.

Imaging and laboratory evaluation

- **National Comprehensive Cancer Network (NCCN), AUA and EAU all recommend computed tomography (CT) of abdomen and pelvis with and without contrast, with excretory phase** (unless contrast is contraindicated). Purposes: nodal and metastatic staging, **upper tract assessment**, and local extent and **resectability**.
- **Image before TUR:** post-resection inflammation **mimics T3 disease**.
- **Magnetic resonance imaging (MRI) or magnetic resonance urography (MRU):** for **poor renal function or iodinated contrast allergy**. 3-tesla magnets, **dynamic contrast enhancement (DCE) and diffusion-weighted imaging (DWI)** give **better local T-staging than CT**; MRI may also predict grade and chemosensitivity.
- **Vesical Imaging-Reporting and Data System (VI-RADS)** standardises bladder MRI reporting; **a score of 3 or more is suspicious for MIBC**; sensitivity and specificity for muscle invasion are both high (roughly 80–90%). Requires an **experienced radiologist** and **pre-TUR imaging**; expected to do for bladder cancer what PI-RADS does for prostate cancer.

TABLE 131.1 VESICAL IMAGING REPORTING AND DATA SYSTEM (VI-RADS) SCORE SYSTEM

Category	T2-weighted imaging	DCE	DWI
1 — invasion unlikely	<1 cm , with or without stalk; low-signal muscularis propria line intact	No early enhancement of muscularis propria	Muscularis propria continuous, intermediate signal
2 — invasion unlikely	>1 cm with stalk and high-signal thickened inner layer, or sessile ; muscle line intact	Early enhancement of inner layer only	As category 1
3 — equivocal	Any size, no category 2 features ; muscle line intact	No interruption or early enhancement of muscularis propria	As category 1
4 — invasion likely	Interruption of the low-signal line	Tumour enhancement extends focally into muscularis propria	High signal
5 — invasion highly likely	Extension into extravesical fat	Enhancement through entire wall and into fat	High signal through entire wall and fat

Adapted from Panebianco V, Narumi Y, Altun E, et al., Eur Urol 2018.

- **What the table tests:** the integrity of the **low-signal muscularis propria line** — intact in 1–3, interrupted in 4, breached into fat in 5.
- **Trap:** the positivity threshold is ≥ 3 , so the "equivocal" category already counts as suspicious.

STAGING MODALITIES COMPARED

Modality	Use	Strength	Weakness
CT urography (with and without contrast)	All patients	Perivesical fat invasion (pT3b), hydronephrosis (strongly associated with extravesical disease), invasion of adjacent organs, upper tracts; accuracy rises with stage	Nodal sensitivity poor (roughly 30–50%); post-TUR artefact
MRI / VI-RADS	Renal impairment, contrast allergy; local staging	Best local T-staging	Expertise-dependent; must precede TUR
¹⁸F-fluorodeoxyglucose (FDG) positron emission tomography (PET)/CT	Not routine; AUA: when conventional imaging is abnormal and biopsy is not feasible	Modestly better than CT for nodes; alters management in a meaningful minority (mostly curative → palliative)	Useless for local staging (urinary FDG excretion); ¹¹ C-choline, -acetate and -methionine add little
Chest imaging	All MIBC	Chest CT preferred over radiograph in current or former smokers	—
Bone and brain imaging	Not routine (metastases rare)	Bone: MRI, PET/CT or bone scan for bone pain or raised alkaline phosphatase ; brain: only for neurological signs	—

- **Blood tests: comprehensive metabolic panel (CMP)** — liver function, electrolytes, renal function — and **complete blood count (CBC)** for anaemia. **The CMP determines eligibility for NAC and orthotopic diversion.**

Concept — Every clinical staging tool errs towards understaging, and understaging costs lives: hence imaging before resection, complete and deep TUR, frozen sections at surgery, and systemic therapy aimed at disease no scan can see.

PATHOLOGIC STAGING

- **American Joint Committee on Cancer (AJCC) tumour–node–metastasis (TNM) system, 8th edition,** used both clinically (cTNM) and pathologically (pTNM).

TABLE 131.2 TUMOR NODE METASTASIS STAGING SYSTEM FOR BLADDER CANCER

Category	Definition
Ta	Noninvasive papillary carcinoma
Tis	Carcinoma in situ — " flat tumour "
T1	Invades lamina propria (subepithelial connective tissue)
T2a / T2b	Invades muscularis propria — inner half / outer half
T3a / T3b	Invades perivesical tissue — microscopically / macroscopically (extravesical mass)
T4a	Invades prostatic stroma, seminal vesicles, uterus, vagina
T4b	Invades pelvic wall, abdominal wall
N1	Single regional node in the true pelvis (obturator, internal and external iliac, perivesical , presacral)
N2	Multiple regional nodes in the true pelvis
N3	Common iliac nodes

Category	Definition
M1a	Nodal-only distant metastasis — nodes beyond the common iliacs
M1b	Non-nodal distant metastasis
Stage 0a / 0is / I	Ta N0 M0 / Tis N0 M0 / T1 N0 M0
Stage II	T2a–T2b N0 M0
Stage IIIA	T3a–T4a N0 M0, or T1–T4a N1 M0
Stage IIIB	T1–T4a N2–N3 M0
Stage IVA	T4b any N M0, or any T any N M1a
Stage IVB	Any T any N M1b

Regional nodes include primary and secondary drainage regions; all other nodes above the aortic bifurcation are distant. From American Joint Committee on Cancer, TNM staging system for bladder cancer, 8th edition, 2018.

- **Substaging is prognostic: pT2a outperforms pT2b, and pT3b does worse than pT3a**, for recurrence-free and cancer-specific survival — the reason both substages are retained.
- **Prostatic invasion (pT4a) requires established stromal invasion** — directly through the bladder neck, or posteriorly via the seminal vesicles and periprostatic ducts. **CIS of the prostatic urethra or ducts does not upstage.**
- **Organ-confined disease = pT2bN0M0 or less.**
- **8th-edition nodal changes: perivesical nodes are now regional (N1); nodes beyond the common iliacs are M1a**, because nodal-only distant disease survives better than non-nodal metastasis (M1b).
- **Traps:** common iliac involvement is **N3, not M1**; stage IIIA already includes **N1**; **T4b is stage IVA even when N0.**

Staging Pitfalls

- **High-grade T1: repeat resection within 6 weeks** of the initial transurethral resection of bladder tumour (TURBT), because upstaging is common.
 - **No muscularis propria in the specimen → upstaging on re-resection in over 40%**; even with muscle present, **about a fifth to a third are upstaged to MIBC.**
 - AUA and EAU: **repeat TUR for T1 disease without muscle in the specimen** (restaging TUR: Chapter 130).

Molecular Classifications

- **Four systems:** University of North Carolina (UNC), MD Anderson Cancer Center (MDA), The Cancer Genome Atlas (TCGA) and Lund University — derived mainly from MIBC cohorts, each with **three to five subtypes** (e.g. UNC: claudin-low, basal, luminal; MDA: basal, p53-like, luminal).
- **The systems are concordant:** MDA basal ≈ Lund squamous cell subtype; MDA p53-like ≈ Lund infiltrated (both **extracellular matrix-enriched**); MDA luminal ≈ Lund urobasal A (both carry **activating fibroblast growth factor receptor 3 [FGFR3] mutations**).

MOLECULAR SUBTYPES: BEHAVIOUR AND CHEMOSENSITIVITY

Subtype	Features	Response to NAC	Survival
Luminal	FGFR3 activating mutations	Responds	Best overall survival (OS), with or without NAC
Basal	Sarcomatoid features, squamous differentiation, metastatic at presentation	Responds — largest survival gain from NAC	Shorter OS and disease-specific survival (DSS)

Subtype	Features	Response to NAC	Survival
p53-like	Extracellular matrix-enriched	Resistant	—

- **Subtype-directed treatment is not yet standard**; future targeted therapy may follow each subtype's profile.

Circulating Tumor Markers

- **Imaging remains the standard** for assessing disease burden; **circulating tumour DNA (ctDNA)** is a promising alternative.
- **ctDNA clears during neoadjuvant and adjuvant immunotherapy** and tracks **overall prognosis**.
- In cisplatin-ineligible patients given neoadjuvant immunotherapy, ctDNA positivity **fell stepwise from baseline → after immunotherapy → after surgery**; **preoperative ctDNA predicted higher T and N stage**; **no ctDNA-negative patient relapsed**, and most relapses were ctDNA-positive.
- **Over a third remain ctDNA-positive after RC**, with poor prognosis; in the IMvigor010 exploratory analysis, **adjuvant atezolizumab increased ctDNA clearance and improved disease-free and overall survival in ctDNA-positive patients**.
- ctDNA may become a **prognostic and predictive** decision aid.

Concept — Pathological stage is prognosis: organ confinement ($\leq pT2b$) and nodal status carry most of the information, and molecular subtype and ctDNA are emerging as the tools that will say who needs more than surgery.

TREATMENT

- **Discuss all curative options in a multidisciplinary setting** before treatment — urologist and medical oncologist, plus a radiation oncologist if bladder preservation is considered — weighing disease factors, comorbidity and patient goals.
- **Do not delay: a delay of more than 12 weeks (3 months) from diagnosis to RC worsens outcome** — more extravesical tumours and nodal metastases, poorer disease-specific and overall survival — consistently across surgical series, population data (roughly doubled disease-specific and overall mortality) and meta-analysis.

RADICAL CYSTECTOMY AND BILATERAL PELVIC LYMPH NODE DISSECTION

- **RC with bilateral pelvic lymph node dissection (PLND) is the standard of care** for localised, resectable MIBC.
- Most comparative studies show **poorer survival with bladder preservation**, but those patients are **older and more comorbid**, and **quality of life was not assessed**.
- About **80% of MIBC patients having definitive treatment undergo RC**.
- **Excellent local control:** pelvic recurrence **as low as about 4% in node-negative disease**.
- **Randomised data show superior outcomes with NAC** (see Neoadjuvant Chemotherapy).

EXTENT OF RADICAL CYSTECTOMY

Males	Females (classical anterior exenteration)
Bladder and perivesical soft tissue, prostate, seminal vesicles	Bladder, urethra, uterus with cervix, ovaries, fallopian tubes, anterior vaginal wall

- Growing attention to urinary and sexual quality of life has driven **organ preservation in both sexes** — only for **organ-confined disease without bladder neck, urethral or prostatic involvement**.

Organ-sparing cystectomy in males

- **Options:** preservation of the **neurovascular bundles**, the **prostatic capsule or entire prostate**, and the **seminal vesicles**.
- **Prostate-sparing requires:** no bladder neck disease + negative prostatic urethral biopsy + negative prostate biopsy.
- **Why caution:** incidental **prostate cancer in roughly a quarter to a half of cystoprostatectomy specimens** (up to a third clinically significant), and **UC involving the prostate** in a substantial proportion; predictors are **bladder neck tumour, multifocality and CIS**.
- **Evidence is small retrospective series:** function tracks the **amount of tissue spared**; apical-, posterior- and total-sparing techniques all report good continence and erectile function.
- **Capsule-sparing versus nerve-sparing:** a small randomised trial found **no clinically significant difference** in oncological or functional outcome; a retrospective comparison found capsule- and seminal-vesicle-sparing gave **comparable oncology and better sexual function — conflicting**.
- Reserve for **carefully selected men** who wish to preserve sexual function and fertility.

Organ-sparing cystectomy in females

- **UC involving the gynaecological organs is uncommon (about 5–6%).**
- **Complete urethrectomy can be omitted unless the bladder neck is involved**, permitting orthotopic substitution.
- **No tumour at the bladder base or trigone → uterus and anterior vagina may be spared:** better neobladder support, **less urinary retention**, preserved autonomic nerves and **less postoperative pelvic organ prolapse**.
- **Premenopausal ovarian preservation** in women without breast or ovarian cancer risk maintains hormonal homeostasis. **Increased endometrial, breast or ovarian cancer risk → salpingo-oophorectomy** in women beyond childbearing and older than 40.
- In selected patients: **similar oncological control, better function**.
- **Counsel both sexes about the impact of RC on sexual function.**

Concept — An organ may be spared only where it is oncologically silent: in men the gatekeeper is the prostatic urethra and bladder neck, in women the bladder neck, base and anterior vaginal wall.

BILATERAL PELVIC LYMPH NODE DISSECTION

- **A meticulous PLND is essential.** About 25% have nodal metastases at RC, and **nodal status is the most powerful surrogate for long-term recurrence-free and overall survival**.
- **Node-positive:** about 70–80% recur, versus about 30% after a negative PLND.
- **Extent of PLND independently predicts survival and local recurrence**, even adjusting for chemotherapy and pathology. What constitutes an adequate dissection, and its therapeutic value, is less clear.

TABLE 131.3 INCIDENCE OF PATHOLOGIC PELVIC NODE METASTASIS (%) AT RADICAL CYSTECTOMY IN SELECTED CONTEMPORARY SERIES (2000–2012)

Pathological T stage	Node-positive at RC (range across series)
≤pT1	About 1–7%
pT2	About 8–26% (pT2a lower than pT2b)
pT3	About 33–50%
pT4	About 36–80% (highest in pT4b)

Data from seven contemporary surgical series, 2000–2012.

- **Nodal risk climbs with every T stage**, roughly doubling from organ-confined to extravesical disease.
- **Trap:** even $\leq$ pT1 disease carries a small nodal risk.

Anatomic Extent of Pelvic Lymph Node Dissection

- **Primary drainage:** internal iliac, external iliac, obturator and presacral nodes. **Secondary drainage:** common iliac, para-aortic, interaortocaval and paracaval nodes.
- **Standard PLND = bilateral external iliac, internal iliac and obturator nodes.**

PELVIC LYMPH NODE DISSECTION TEMPLATES

Border	Standard dissection	Extended dissection
Medial	Ureter	Ureter
Lateral	Genitofemoral nerve	Genitofemoral nerve
Caudal	Inguinal ligament / node of Cloquet	Same
Cranial	Ureter crossing the common iliac, or common iliac bifurcation	Aortic bifurcation, up to the inferior mesenteric artery (IMA) — controversial

- **Nodal mapping series:**
 - A "super-extended" dissection to the IMA found nodes outside the true pelvis **only with multiple positive pelvic nodes, and no skip metastases** → the **obturator and internal iliac nodes are the sentinel drainage**; a negative frozen section there made higher dissection unnecessary.
 - Standard (to the iliac bifurcation) versus extended (to the aortic bifurcation): more nodes retrieved, but **no staging advantage** — the same proportion node-positive; para-aortic positives always accompanied lower positives.
- **Skip metastases are rare, but common iliac nodes are involved in about 7–13% → consider removing the common iliac nodes.**

Number of Lymph Nodes Identified at the Time of Cystectomy

- **Higher yield → higher sensitivity** for nodal metastasis; "adequate" yields quoted as **12–25 nodes**.
- **Modelling:** removing 25 nodes gives about a 75% chance of detecting nodal disease (about 50% with 15–25 nodes, 90% with 45) → **25 nodes as a reasonable staging minimum**.
- **Node count carries prognostic and staging information in both node-positive and node-negative patients.**
- **A therapeutic threshold is unproven** (no prospective data; most studies cite **9–16 nodes**): in a Southwest Oncology Group (SWOG) 8710 reanalysis, **fewer than 10 nodes doubled the risk of death**; another series linked **16 or more nodes** to better survival. **Ten nodes is the commonest threshold for pelvic recurrence.**
- **Node count reflects surgery and pathology alike** — template extent and the **number of packets submitted**.

Lymph Node Density and Extracapsular Nodal Extension

- **Lymph node density** = positive nodes ÷ total nodes removed.
- Early series: **density below 20% → better DSS**, and better response to adjuvant chemotherapy; other studies found **no advantage over simple node status**; a meta-analysis linked density to poorer DSS, but with **marked heterogeneity**.

- **Extranodal extension** (present in about half of node-positive cases) **independently predicts recurrence, cancer-specific mortality and overall survival.**

Pelvic Lymph Node Dissection and Oncologic Outcomes

LIMITED VERSUS EXTENDED DISSECTION — THE EVIDENCE

Study	Design and population	Nodes (limited vs extended)	Result
Two-centre consecutive series	Retrospective; limited to common iliac bifurcation vs extended to ureter crossing	—	Extended: better 5-year recurrence-free survival, most marked in pT3
German phase III trial	Randomised; limited (obturator, internal and external iliac) vs extended to IMA; T1 included, NAC excluded	Median 19 vs 31	No benefit in recurrence-free (59% vs 65%, not significant), cancer-specific or overall survival
SWOG S1011	Randomised; pT2–4a, most received NAC — mirrors current practice; median follow-up 6 years	Median 24 vs 39; node-positive rate similar	No difference in disease-free survival (HR 1.10) or OS (HR 1.15); more grade ≥3 adverse events at 90 days with extended

- **Verdict: standard PLND is the standard of care;** extended dissection adds morbidity without survival benefit.

Concept — Nodes are a staging problem more than a therapeutic one: removing more nodes changes what is counted, not who survives — both randomised trials found no benefit from extended dissection.

INTRAOPERATIVE DECISION MAKING

Grossly Positive Nodes and T4b Disease

- **Clinically node-positive before RC** → cisplatin-based chemotherapy first.
- **Radiographic complete or partial response** → **evaluate for consolidative RC:** pathological complete response in about 14–25%, and complete responders have **markedly better cancer-specific survival;** responders who forgo surgery **mostly recur** (data preceding maintenance immunotherapy). **Strongly consider consolidative RC** in fit responders.
- **Unexpected lymphadenopathy at RC:** frozen section to confirm → proceed to **extended PLND and RC** only if bladder and nodes are completely resectable and the nodes lie within the true pelvis.
- **Do not proceed with RC if:** nodes are **unresectable;** **extensive periureteral disease;** bladder **fixed to the pelvic sidewall;** invasion of the rectosigmoid colon.
- **Aborted RC** → **very poor prognosis:** most die of progression within about 2 years; no 5-year survivors in one series.

Intraoperative Frozen Sections of the Ureter

- **Distal ureteric involvement at RC:** about 6–8%; the **final ureteric margin independently predicts upper tract recurrence (UTR).**
- **UTR is uncommon** — about 4–10% — occurs mostly **2–4 years after RC,** and is often **locally advanced and worse than de novo upper tract UC.**
- **UTR risk factors:** positive ureteric margin, prior bacille Calmette-Guérin (BCG), bladder CIS, preoperative hydronephrosis.

- **Controversy: neither AUA nor EAU gives a recommendation** on intraoperative ureteric margins; a negative margin has not consistently improved outcome.
- **Frozen section accuracy:** sensitivity roughly 70–77%, specificity 83–96%; a positive frozen section makes a positive final margin far more likely.
- **Sequential resection:** most initially positive margins can be cleared, but **"converted" margins still carry higher UTR risk** than initially negative ones, and **persistently positive margins the highest**; failure to convert predicts worse cancer-specific mortality.
- **Skip lesions** (negative distal frozen section, positive proximal final margin) occur **predominantly with CIS**.
- **Practice:** clear the distal ureter when feasible; **frozen section whenever gross tumour is encountered**; for **CIS alone**, resect maximally **without compromising length for the diversion** — the benefit of a CIS-negative margin is debatable given skip lesions.

Prostatic Urothelial Carcinoma and Management of the Distal Urethra

- **Male urethral recurrence after RC: 1–8%**; most are symptomatic; **most occur within 2 years** (median about 8–28 months); periodic cytology helps in high-risk men.
- **Risk factors:** prostatic urethral involvement, **prostatic stromal invasion (highest risk — up to about 30%)**, bladder neck tumour, CIS, multifocality; stromal invasion outranks prostatic urethral CIS and ductal or acinar involvement.
- **Urethrectomy:** consider for **diffuse CIS of the prostatic urethra or prostatic stromal invasion**; the authors perform **en bloc urethrectomy for invasive prostatic urethral disease**; some reserve it for a positive apical urethral margin.
- **A positive preoperative prostatic urethral biopsy does not preclude orthotopic diversion:** take an intraoperative frozen margin, and if negative, orthotopic diversion may proceed.
- **Frozen urethral margin:** specificity near 100%, sensitivity variable (about 33–93%). **Perform it in any patient considered for orthotopic diversion.**
- **Small low-grade papillary urethral tumours:** resect before RC and **retain the urethra** if there is no other indication for urethrectomy.
- **Lower urethral recurrence after orthotopic diversion** is more likely **selection bias** than a protective effect of urine exposure (no physiological evidence).

Managing the Female Urethra

- **Only T4 tumours involving the urethra and/or vagina mandate complete urethrectomy** (no urethral preservation).
- **Urethral involvement: about 2–12%.** Risk features: primary tumour at the **bladder neck, posterior bladder base or anterior vaginal wall**, or inguinal lymphadenopathy.
- **About 60% of bladder neck tumours have no urethral tumour on final pathology**, so urethrectomy is not automatic.
- **Proximal urethral frozen section correlates well with the final margin — perform in every woman considered for orthotopic substitution.**
- **Isolated urethral recurrence after orthotopic substitution: about 1%.**

URETHRAL MANAGEMENT: MALES VERSUS FEMALES

	Males	Females
Frequency	Urethral recurrence 1–8%	Urethral involvement 2–12%; recurrence after neobladder about 1%

	Males	Females
Highest-risk feature	Prostatic stromal invasion	Bladder neck, posterior bladder base, anterior vaginal wall
Urethrectomy	Consider for diffuse prostatic urethral CIS or stromal invasion	Mandatory for T4 involving urethra/vagina
Gatekeeper for orthotopic diversion	Intraoperative frozen urethral margin	Proximal urethral frozen section

Concept — *The intraoperative frozen section, not the preoperative biopsy, decides orthotopic diversion — the preoperative biopsy only sets the level of suspicion.*

ONCOLOGIC OUTCOMES AFTER RADICAL CYSTECTOMY

- **Pathological T stage and nodal status are the strongest predictors of progression and survival.**

TABLE 131.4 PERCENTAGE OF 5-YEAR DISEASE-SPECIFIC SURVIVAL (DSS) BY PATHOLOGIC STAGE AFTER RADICAL CYSTECTOMY WITH AND WITHOUT PELVIC LYMPH NODE METASTASIS: SELECTED SERIES REPORTING DSS (2000–2012)

Pathological stage	5-year DSS — table (series ranges)	5-year DSS — body text
pT0 / non-invasive / ≤pT1	About 76–90%	Approaching 90%
pT2	About 70–81%	70–81% (pT2a better than pT2b)
pT3	About 40–53%	40–52%
pT4	About 16–44%	—
Node-negative	About 62–80%	—
Node-positive	About 21–35%	10–35%
Population data by extent	Localised 72%, regional 51%, distant 10%	—

Data from seven surgical and population series, 2000–2012. The body text's lower node-positive bound (10%) includes the population series' distant-disease figure, which the node-positive surgical column does not.

- **pT3 substaging** discriminates in **node-negative** disease (pT3a better recurrence-free and disease-specific survival) but **not in node-positive** disease. One series found pT3b patients without neoadjuvant chemotherapy had poorer metastasis-free survival, whereas pT2b–3a N0 disease matched pT3b disease given adjuvant chemotherapy.
- **Node-positive disease: poor prognosis, yet long-term survival is possible with low-volume nodal disease;** much of the data predates adjuvant immunotherapy and may **underestimate survival**.
- **Positive soft-tissue margin** (about 6% of RC) **raises recurrence and cancer-specific mortality** (hazard ratio [HR] about 1.5 for each).
- **Other prognostic factors:** LVI, molecular markers, **surgeon expertise and hospital volume**, body mass index and age.
- **Despite aggressive surgery, about 50% of RC patients ultimately die of bladder cancer** — usually metastatic recurrence within the first 2 years.

PELVIC (LOCAL) RECURRENCE RISK GROUPS AFTER RADICAL CYSTECTOMY

Risk group	Definition	5-year pelvic failure
Low	≤pT2	About 5–8%
Intermediate	pT3–4, negative margins and ≥10 nodes	About 15–21%

Risk group	Definition	5-year pelvic failure
High	pT3–4 with positive margins or <10 nodes	About 39–46%

- **Predictors of pelvic recurrence: pT3–4 disease, positive soft-tissue margin, fewer than 10 nodes removed;** the model holds across geographically and temporally diverse cohorts.

Concept — Outcome after cystectomy is set by **what lay outside the bladder** — extravesical extension, nodes and margins — which is exactly the population that neoadjuvant, adjuvant and radiation strategies target.

NEOADJUVANT THERAPY FOR MUSCLE-INVASIVE BLADDER CANCER

Neoadjuvant Chemotherapy

- **RC alone is insufficient: nearly half progress to metastatic disease** after cystectomy; cisplatin-based chemotherapy has produced responses since the mid-1980s.
- **Offer NAC only to cisplatin-eligible patients; cisplatin-ineligible patients proceed directly to definitive locoregional therapy.**

TABLE 131.5 PLATINUM ELIGIBILITY

Criterion	Cisplatin-ineligible (any one)	Platinum-ineligible
Renal function	Glomerular filtration rate (GFR) <60 mL/min	Creatinine clearance (CrCl) <30 mL/min
Eastern Cooperative Oncology Group (ECOG) performance status	>1 (i.e. ≥2)	≥3 in the table; ≥2 in the body text
Hearing	Grade ≥2 audiometric hearing loss	—
Peripheral neuropathy	Grade ≥2	Grade ≥3
Heart failure	New York Heart Association (NYHA) class III–IV	NYHA class III–IV
Other	—	Multiple high-grade comorbidities (body text)

- **What the table tests: any single criterion** makes a patient cisplatin-ineligible — **nearly half of bladder cancer patients.**
- The table and body text differ on the performance-status threshold for platinum ineligibility; both are shown.
- **Advantages of NAC:** better tolerated before surgery (**higher compliance**) → treats **micrometastases while burden is low** → **downstages** bulky and locally advanced tumours (more negative margins). **Downstaging to non-invasive disease or pT0 predicts far better outcome** — 5-year recurrence-free survival about 90% for pT0.
- **Disadvantage:** delays definitive local therapy in **non-responders**, who may progress during NAC.

LANDMARK NEOADJUVANT CHEMOTHERAPY EVIDENCE

Trial	Neoadjuvant arm	Control	Key result	Verdict and limitations
BA06 30894 (European Organisation for Research and Treatment of Cancer/Medical Research Council [EORTC/MRC]) — largest trial	3 cycles CMV (cisplatin, methotrexate, vinblastine)	RC and/or radiotherapy	At 8 years: cancer-specific mortality HR 0.74; 16% relative reduction in mortality; 10-year OS 36% vs 30%; median survival +7 months	Positive ; CMV is no longer standard, and many received radiotherapy rather than RC
SWOG 8710	3 cycles MVAC (methotrexate, vinblastine, Adriamycin [doxorubicin], cisplatin)	RC alone	Median survival 77 vs 46 months; pT0 38% vs 15%; 5-year OS 57% vs 43% (P = .06); pT0 → 5-year survival 80% vs 40% with residual disease	Trend, not significant ; secondary analysis: benefit in mixed histology (squamous or glandular differentiation)
VESPER (GETUG-AFU V05)	6 cycles dose-dense MVAC (ddMVAC)	4 cycles gemcitabine-cisplatin (GC)	Organ-confined response (<ypT3N0) 77% vs 63%; 3-year progression-free survival 66% vs 56% (neoadjuvant cohort)	ddMVAC superior , at the cost of more asthenia and gastrointestinal toxicity
Advanced Bladder Cancer (ABC) meta-analysis (11 randomised trials)	Platinum combination	No NAC	OS HR 0.86 — 5% absolute benefit at 5 years (45% → 50%); disease-free survival HR 0.78	Combination platinum only ; single-agent platinum no benefit
Latest meta-analysis (15 randomised trials)	NAC	No NAC	8% absolute survival improvement	Confirms benefit

TABLE 131.6 RANDOMIZED TRIALS OF NEOADJUVANT CHEMOTHERAPY

Trial	Neoadjuvant regimen	Local therapy	Survival result
EORTC/MRC (BA06)	CMV	RC or radiotherapy	10-year OS 36% vs 30% (P = .037) — positive
SWOG 8710	MVAC	RC	5-year OS 57% vs 43% (P = .06)
Abol-Enein	Carboplatin, methotrexate, vinblastine	RC	5-year OS 62% vs 42% (P = .013) — positive
Nordic I	Cisplatin + doxorubicin	Radiotherapy + RC	5-year OS 59% vs 51% — not significant
Nordic II	Cisplatin + methotrexate	RC	5-year OS 53% vs 46% — not significant
GUONE (Italy)	MVAC	RC	5-year OS 55% vs 54% — not significant
GISTV (Italy)	MVEC (methotrexate, vinblastine, epirubicin, cisplatin)	RC	Median OS — not significant
CUETO (Spain)	Single-agent cisplatin	RC	6.5-year OS 35% vs 37% — not significant
WMURG/ABCSG	Single-agent cisplatin	Radiotherapy	No difference

Trial	Neoadjuvant regimen	Local therapy	Survival result
DAVECA 89-02	Cisplatin + methotrexate	Radiotherapy	5-year OS 19% vs 24% — not significant
Coppin	Cisplatin	Radiotherapy or RC	3-year OS 47% vs 33% — not significant

The source marks the trials of most importance in bold; the EORTC/MRC and SWOG 8710 trials are shown in bold here.

- **Pattern:** the **single-agent cisplatin trials were negative**; benefit belongs to **cisplatin-based combinations** — the conclusion the ABC meta-analysis formalised.
- **Trap:** SWOG 8710's overall-survival difference was **not statistically significant (P = .06)**, despite the large median-survival gap.
- **Other regimens (phase II):** four cycles of **ddMVAC** → pathological response below pT1N0M0 in about half; three cycles of "**accelerated**" **MVAC** → pT0 in over a third; both well tolerated.
- **GC** was adopted from the metastatic randomised trial showing **equal efficacy and less toxicity than MVAC**; retrospective neoadjuvant data show **no survival difference between GC and ddMVAC**.
- **Guidelines (AUA, EAU, NCCN):** **cisplatin-based NAC for all cT2–cT4aN0M0**. NCCN: **ddMVAC for 3–6 cycles preferred**; **GC for 4 cycles the alternative**. In fit patients with few comorbidities, **ddMVAC may be superior**.
- **Cisplatin-ineligible:** **NAC not recommended**; **carboplatin regimens show no OS benefit** → proceed to locoregional therapy.

Neoadjuvant Immunotherapy and Chemoimmunotherapy Combinations

- **Programmed death-1 (PD-1)** on activated T cells binds **programmed death ligand-1 (PD-L1)**; tumour cells express PD-L1 to **neutralise T-cell activation**. **Cytotoxic T lymphocyte-associated antigen 4 (CTLA-4)** on activated T cells engages receptors on antigen-presenting cells to **inhibit T-cell proliferation**. Blocking either **increases T-cell activation and proliferation** → antitumour response.

IMMUNE CHECKPOINT INHIBITORS BY TARGET

Target	Location	Agents
PD-L1	Tumour cell (ligand)	Atezolizumab, durvalumab, avelumab
PD-1	Activated T cell (receptor)	Pembrolizumab, nivolumab
CTLA-4	Activated T cell (priming phase)	Ipilimumab, tremelimumab

Figure 131.2 — Mechanism of immune checkpoint inhibition

- **Priming phase:** the dendritic cell presents antigen (major histocompatibility complex [MHC] II to T-cell receptor [TCR]); **CTLA-4 binding CD80/86 acts as an "active brake"**, stopping the response (immunodepression) → **anti-CTLA-4 releases it**.
- **Effector phase:** **PD-1 on the T cell binding tumour PD-L1 blocks tumour killing** → **anti-PD-1/PD-L1 restores it**.
- Source: Wang DR et al., Signal Transduct Target Ther 2022.
- **Established roles of PD-1/PD-L1 inhibition** (see Management of Metastatic Bladder Cancer): metastatic or unresectable UC, second line after chemotherapy, and cisplatin-ineligible patients. Neoadjuvant use is under study, alone or with cisplatin chemotherapy.

NEOADJUVANT IMMUNOTHERAPY AND CHEMOIMMUNOTHERAPY TRIALS

Trial	Regimen	Population	Result
ABACUS (phase II, single arm)	Atezolizumab ×2	Cisplatin-ineligible	Pathological complete response 31%; 2-year disease-free survival 68%, OS 77%; PD-L1 and tumour mutational burden did not predict recurrence-free survival
PURE-01 (phase II)	Pembrolizumab ×3	UC-predominant	pT0 42%; <pT2 54%
Phase II combinations	Pembrolizumab, atezolizumab, durvalumab or nivolumab + GC	Cisplatin-eligible (one also ineligible)	Pathological response (<pT2N0 or ≤ypT1N0) roughly 55–70%; pT0 30–44% across all
NIAGARA, KEYNOTE-866, ENERGIZE (phase III, ongoing)	Durvalumab + GC; pembrolizumab + GC; nivolumab + GC ± indoleamine 2,3-dioxygenase 1 (IDO1) inhibitor	—	Pathological response and event-free survival awaited

- **Limitations:** non-randomised, single-arm, **not designed to compare with chemotherapy** — but promising for cisplatin-ineligible patients.
- **Current position:** insufficient data to recommend neoadjuvant immunotherapy, alone or with chemotherapy.

Neoadjuvant Radiation Therapy

- Common in the 1970s–80s; abandoned after cisplatin-based NAC. Small randomised trials showed **pathological downstaging but no survival benefit → do not offer preoperative radiation.**

Concept — Neoadjuvant benefit belongs to cisplatin combinations and nothing else: carboplatin, single-agent cisplatin, radiation and — so far — immunotherapy have not earned it, and pT0 at cystectomy is the surrogate that shows it working.

ADJUVANT THERAPY FOR MUSCLE-INVASIVE BLADDER CANCER

- **Target population:** pT3–4 or node-positive at RC — high recurrence risk; the aim is to treat micrometastatic disease.

NEOADJUVANT VERSUS ADJUVANT CHEMOTHERAPY

	Neoadjuvant	Adjuvant
Selection	Clinical stage (understaging risk)	True pathological stage — yet still cannot identify who needs it, so overtreatment is guaranteed
Delivery	Better tolerated, higher completion	Post-RC morbidity → delay or non-receipt (a quarter never received it in one trial)
Local therapy	Delayed; non-responders may progress	Immediate; no delay for chemoresistant tumours
Evidence	Level 1 (randomised trials, meta-analysis)	Weaker — small, poorly accruing, early-closed trials

Adjuvant Chemotherapy

TABLE 131.7 ADJUVANT SYSTEMIC THERAPY TRIALS AFTER CYSTECTOMY — CHEMOTHERAPY

Trial	Regimen	Result	Shortcomings
USC trial (first randomised)	Cisplatin-based (CISCA-like, single-agent cisplatin, or assay-selected) vs observation, pT3–4 or N+	Body text: 3-year progression-free 70% vs 46%; median survival 4.3 vs 2.4 years (P = .0062). Table: labels 4.3 vs 2.4 years as disease-free survival	Small; regimens unstandardised; 25% randomised to chemotherapy received none
German trial	MVAC or MVEC ×3 vs observation, pT3b–4a and/ or N+	3-year disease-free survival 63% vs 13%; 10-year progression-free 44% vs 13%, DSS 42% vs 17%; no OS benefit	Small; closed early ; only 69% received chemotherapy; no chemotherapy at relapse in controls
CMV trial	CMV	Longer time to progression	Closed early; no OS benefit
CISCA series	CISCA	Benefit	Not randomised
Cisplatin; CM; MVEC trials	Single-agent cisplatin; cisplatin + methotrexate; MVEC	No benefit (CM: none for N0)	Small
USC/SWOG p53 trial	MVAC ×3, pT1–2 N0 p53-positive	No recurrence-free benefit	Stopped at interim analysis
Italian phase III	GC ×4 , pT2G3–pT4, N0–2	No OS or disease-free survival difference (5-year OS about 48% overall)	Closed early for poor accrual
CALGB, EORTC 30994, SOGUG 99/01	Various	—	All closed for poor accrual

CISCA, cisplatin, cyclophosphamide, doxorubicin (Adriamycin); CM, cisplatin and methotrexate; CALGB, Cancer and Leukemia Group B; SOGUG, Spanish Oncology Genitourinary Group. For the USC trial the body text reports median survival where the table reports disease-free survival; both are shown.

- **ABC meta-analyses:** 2005 (6 trials): **OS HR 0.75 — a 25% relative reduction in death.** 2022 individual-patient update (10 trials): **cisplatin-based adjuvant chemotherapy improves OS by 6% at 5 years (9% after adjustment)**, plus recurrence-free, locoregional recurrence-free and metastasis-free survival.
- **Greatest benefit: node-positive and extravesical disease; no benefit shown in variant histology.**
- **Verdict:** not standard of care because of the trials' deficiencies, but **for cisplatin-naïve patients with pT3–4 or node-positive disease, discuss adjuvant ddMVAC or GC.**

Adjuvant Immunotherapy

TABLE 131.7 ADJUVANT SYSTEMIC THERAPY TRIALS AFTER CYSTECTOMY — IMMUNOTHERAPY

Trial	Agent	Population	Result
CheckMate 274 (phase III, double-blind)	Nivolumab vs placebo, 12 months	pT3, pT4a or N+ after RC	Median disease-free survival 20.8 vs 10.8 months → FDA approval for high-risk disease irrespective of prior NAC
IMvigor010 (phase III)	Atezolizumab vs observation	pT2–4a or N+ after NAC; pT3–4a or N+ without NAC	Negative for disease-free survival; benefit in the ctDNA-positive subgroup

Trial	Agent	Population	Result
AMBASSADOR (ongoing)	Pembrolizumab vs observation	≥pT2 after NAC, or cisplatin-ineligible/declined	OS and disease-free survival awaited

- **Why the discordance:** IMvigor010 included pT2 disease; a drug-specific difference between nivolumab and atezolizumab is also possible.
- **Recommendation:** offer adjuvant nivolumab for ≥pT3 (or node-positive) disease in patients ineligible for, or declining, adjuvant cisplatin chemotherapy, and after NAC with persistent ≥pT3 disease.

Adjuvant Radiation

- **pT3–4 disease: local failure 20–40% at 5 years;** after pelvic failure, outcome is catastrophic — **median survival about 9 months.**
- **Cairo randomised trial (1980s):** adjuvant radiotherapy (two techniques) vs observation after RC with negative margins but high local-failure risk → **better pelvic control and 5-year OS;** generalisability questioned because **80% had squamous cell carcinoma** (outcomes appeared similar by histology).
- **Cairo phase II:** GC ×4 vs **sequential GC ×2 → radiotherapy → GC ×2** for pT3b, grade 3 or node-positive disease (about half UC) → **better 2-year pelvic failure-free survival (96% vs 69%);** disease-free and overall survival higher but not significantly (underpowered); **low late grade ≥3 toxicity.**
- **May be considered for pT3 or node-positive UC;** fields cover the **cystectomy bed and pelvic nodes;** **no strong recommendation** possible.

Concept — Adjuvant therapy has the advantage of true pathological staging but **the disadvantage of weak evidence:** only CheckMate 274 is a clean positive trial, which is why adjuvant nivolumab — not chemotherapy or radiation — carries the firmest recommendation.

MULTIMODAL BLADDER PRESERVATION THERAPY

- **Best candidates:** small solitary tumour, no pelvic lymphadenopathy, no CIS, no hydronephrosis, good bladder function.
- Historically reserved for those unfit for surgery; now a viable option in selected MIBC. **RC carries the second-highest 30-day readmission rate** in Medicare claims, and patients are typically older and comorbid.
- **Two populations:** (1) **high operative risk** from comorbidity and frailty; (2) **surgically fit patients with limited disease, adequate bladder urothelium and function, motivated to keep the bladder.** Careful selection and counselling are essential.
- **Trimodality therapy (TMT) — the best-studied approach = visibly complete TUR + radiosensitising chemotherapy + radiation.** Complete TUR with radiation improves local control; radiation also treats adjacent nodes; chemotherapy sensitises.
- **Concurrent sensitising chemotherapy is superior to radiation alone** — established by two randomised trials and many retrospective studies.

RADIOSENSITISING REGIMENS

Regimen	Evidence and use
Mitomycin C + 5-fluorouracil (5-FU)	Largest randomised trial (BC2001): vs radiation alone → 2-year locoregional disease-free survival 67% vs 48%, without significantly worse late toxicity
Cisplatin-based (± 5-FU or paclitaxel)	Well studied; basis of the MGH and RTOG protocols

Regimen	Evidence and use
Gemcitabine alone	Cisplatin-ineligible; effective (3-year metastasis-free rate about 78%)
Nicotinamide + carbogen	Validated alternative

- **Outcomes:** 5-year cancer-specific survival about 50–84% and OS 36–74%; **salvage cystectomy 10–30%.**
- **Multi-institutional propensity-matched analysis, RC vs TMT: equivalent 5-year metastasis-free survival (about 75%) and cancer-specific survival (81% vs 84%);** salvage cystectomy 13% — limited by retrospective design and selected TMT patients.

RADIATION SCHEDULES FOR TRIMODALITY THERAPY

	Split-course	Continuous-course
Sequence	Induction chemoradiation to about 40 Gy → restaging by imaging and cystoscopy → consolidation to 60–66 Gy if no persistent invasive disease	Full course to 60–66 Gy → endoscopic restaging 3–4 months later
Suited to	Surgically fit patients (immediate salvage RC possible)	Fit and unfit patients
Status	Historical — Massachusetts General Hospital (MGH) and Radiation Therapy Oncology Group (RTOG)	Commonest now; favoured by most trials

- **Hypofractionation: 55 Gy in 20 fractions** (bladder-only fields) is **non-inferior** for local control, OS and toxicity; a BC2001–BCON meta-analysis confirmed no difference in early or late toxicity.
- **TUR quality matters:** visibly complete TUR → better 5- and 10-year OS and **fewer cystectomies (about 11% vs 42%)**, though confounded by selection.
- **Requirements:** multidisciplinary care and **excellent compliance with frequent cystoscopic and imaging surveillance**; counsel about possible future salvage therapy.
- **For surgically fit patients, discuss TMT versus RC with or without perioperative chemotherapy.**
Selection: **clinical T2, no hydronephrosis, no CIS, visibly complete TUR.** Only about **6–19% of MIBC** meets strict "limited burden" criteria. The bladder must be **worth preserving** and the patient motivated — **10–30% eventually need salvage cystectomy** for muscle-invasive recurrence. **Very long-term data (>15 years) are limited;** late radiation effects (**secondary malignancy**) make **young, fit patients with long life expectancy less ideal.**
- **MGH/RTOG split-course experience (six prospective protocols):** complete response at the treatment break 69%; 5- and 10-year OS 57% and 36%, DSS 71% and 65%; **80% of 5-year survivors keep an intact bladder; salvage cystectomy 21%,** mostly after induction for incomplete response; after salvage, 5-year OS 45%.
- **Patients unfit for RC:** less stringent criteria, but **extensive CIS and poor baseline bladder function are relative contraindications;** no accepted definition of unfit — clinical gestalt aided by mortality nomograms; confirm that **life expectancy justifies treatment;** multidisciplinary evaluation and shared decision-making.
- **Population data** show worse OS with chemoradiation than surgery (**HR about 1.4**), but are **confounded by selection** — registry TMT patients are largely surgically unfit (unadjusted 5-year OS about 30%, DSS about 55%).
- **Minimal late grade 3–4 toxicity and favourable quality of life;** a significant minority still requires salvage cystectomy.

TABLE 131.8 MAJOR PROSPECTIVE TRIMODAL BLADDER PRESERVATION STUDIES

Sensitiser	Studies	Radiation dose	Complete response	Salvage cystectomy	Survival
Cisplatin-based (single agent or weekly; ± 5-FU or paclitaxel)	Six studies, including the pooled MGH/RTOG report	44–65 Gy (most about 64 Gy)	60–93% (MGH/RTOG 72%)	13–29% (MGH/RTOG 29%)	5-year OS about 49–59%; MGH/RTOG DSS 64%, 59%, 57% at 5, 10, 15 years
5-FU + mitomycin C (BC2001)	Randomised trial and 10-year update	55 or 64 Gy	Not reported	11–14%	5-year OS 48% (original report); 5-year OS 67.6% and recurrence-free survival 63% (10-year update)

The two BC2001 reports give different 5-year OS figures for the same cohort; both are shown.

- **Pattern:** complete response is common and salvage cystectomy is consistently needed in roughly 1 in 10 to 1 in 4, whichever sensitiser is used.

Concept — Trimodality therapy is a strategy that contains cystectomy, not an alternative to it: salvage cystectomy in 10–30% is the designed safety net, so selection — T2, no CIS, no hydronephrosis, visibly complete TUR — determines success.

SINGLE-MODALITY TREATMENT

Chemotherapy Monotherapy

- Chemotherapy's role is **neoadjuvant or within TMT**; alone it rarely produces durable remission, though some patients are pT0 at cystectomy.
- **Complete responders who declined RC:** at 5 years, about two-thirds alive and half with an intact bladder, but **about a third died of bladder cancer**; long survivors had **small, solitary, low-stage, completely resected tumours**. An MVAC + TUR series: complete response about 50%; 60% alive without further therapy at about 5 years.
- **Intravesical recurrence is common and early** — nearly half at a median of about 5 months in a cT0 series.
- **RETAIN BLADDER (phase II):** surveillance after NAC for patients with **DNA damage repair mutations (ATM, ERCC2, FANCC, RB1)** and no visible disease → **half recurred** at about 15 months, several with MIBC.
- **Not recommended as primary therapy for localised MIBC.**

Radiation Monotherapy

- (Here including debulking TUR before radiation; excludes TMT and adjuvant radiotherapy.) **Inferior to radiation with sensitising chemotherapy; reserve for quasi-palliative situations.**
- Evidence: 5- and 10-year OS about 40% and 19%, inferior to chemoradiation; **local control as low as 27% in T3**; the **BC2001 radiation-alone arm was inferior to chemoradiation**; in patients aged 80 or more, **chemoradiation improved 2-year OS over external beam radiation therapy (EBRT) alone.**
- **Predictors of local failure (as for radical TUR):** tumour size, multifocality, hydronephrosis, CIS, higher clinical T stage, incomplete TURBT.
- **Curative bladder dose: 64–66 Gy in 32 fractions, or 55 Gy in 20 fractions;** treat nodes at the physician's discretion, especially if node-positive.
- **Not routine;** use in patients **unfit for surgery or chemoradiation**, including for **symptom control.**

Radical Transurethral Resection

- **Viable only for tumour confined to superficial muscle.** Rationale: **pT0 in about 10%** of patients treated with RC alone.
- **Limitations: understaging** (pT3 up to about 40% and pT4 about 9% of cT2 cystectomy specimens) and **about 25% occult nodal metastasis.**
- **Long-term series (>10 years):** after restaging TUR showing no tumour or residual NMIBC, **a third developed MIBC, and nearly half of those died;** another cohort: over half recurred.
- **Highly selected contemporary series (tumour <3 cm, no hydronephrosis; intravesical therapy for CIS; re-resection at 3 and 6 months; 3-monthly cystoscopy for 2 years):** cancer-specific survival about 82%, 80%, 77% at 5, 10, 15 years, but **progression-free survival with intact bladder falling to about 58% at 15 years.**
- **Not a primary treatment;** counsel patients who are unfit or decline RC and TMT about the **substantial recurrence risk.**

Partial Cystectomy

- **Associated with higher mortality than RC** (no randomised comparison). **Advantages over radical TUR:** full PLND for staging, and **full-thickness wall with perivesical fat removed.**
- Historically poor (5-year OS as low as 24%; frequent local and **wound recurrence**); acceptable with **strict selection**, when long-term results may approach RC.

PARTIAL CYSTECTOMY — IDEAL CANDIDATE

Requirement

Small (<3 cm), solitary, accessible tumour

No CIS, no hydronephrosis

Broad negative margin achievable, ideally away from the ureteric orifices (avoids reimplantation)

Adequate functional capacity after resection

Routine intraoperative frozen sections; convert to RC if a negative margin cannot be achieved

- **Treatment of choice for urachal adenocarcinoma** — a pathologically distinct entity.
- **Outcomes (UC):** non-muscle-invasive recurrence 12–50%, muscle-invasive 17–57%, metastatic 14–52%; 5-year OS 67–70%, recurrence-free survival 39–62%, DSS 84–87%; intact bladder 65–74%. **Salvage cystectomy in up to a quarter; late recurrence common** (reported beyond 11 years) → **long-term follow-up.**
- **With neoadjuvant chemoradiation** (tumour <25% of bladder area, sparing bladder neck and trigone, complete TUR; 40 Gy EBRT + cisplatin): minimal residual MIBC and excellent 5-year survival in a small series. Also used after NAC in partial and complete responders, and in patients unfit for or refusing RC.
- **Always add PLND; consider perioperative cisplatin chemotherapy.**

SINGLE-MODALITY OPTIONS COMPARED

Modality	Candidates	Chief limitation	Verdict
Chemotherapy alone	Complete responders after NAC; small solitary tumours	Early intravesical recurrence	Not recommended
Radiation alone	Unfit for surgery or chemoradiation	Inferior to chemoradiation	Quasi-palliative; symptom control
Radical TUR	Superficial muscle invasion only	Understaging; occult nodes	Not primary treatment

Modality	Candidates	Chief limitation	Verdict
Partial cystectomy	<3 cm solitary, no CIS, broad margin; urachal adenocarcinoma	Recurrence, including late	Selected patients, always with PLND

Concept — Every single modality fails for the same reason — **it leaves untreated either the residual urothelium or the occult nodes** — which is why each survives only as an option for highly selected or unfit patients.

FOLLOW-UP

- **Goals:** detect **local (pelvic or bladder), distant and urothelial** recurrence; schedules differ after RC and after bladder preservation.
- **Local recurrence:** in the pelvis, **usually within 24 months** (rarely up to 5 years); 5–15% with aggressive disease (high pT, node-positive, positive margins); **poor prognosis**; treated with chemotherapy and radiation.
- **Distant recurrence:** historically **32–70%, mostly within 2 years** — probably lower with modern NAC and adjuvant immunotherapy, though late recurrence occurs; commonest sites **distant nodes, lung, liver, bone**; survival after recurrence is poor.
- **Urothelial recurrence:** **urethra about 4%; upper tract 4–10%**. Urethral risk: tumour in the **prostatic urethra (males) or bladder neck (females)**. **The upper tract is the commonest site of late recurrence**, and most such patients die of metastasis; **biggest risk factor: positive ureteric margin**; detection is best with **upper tract imaging plus cytology**.
- **Whether detecting asymptomatic recurrence improves survival is controversial** — retrospective data conflict; no prospective data.

Follow-Up After Radical Cystectomy

POST-CYSTECTOMY CT SURVEILLANCE BY GUIDELINE

Guideline	CT (contrast, with delayed phase for upper tracts)
EAU	Every 6 months for 3 years , then annually to 5 years
AUA	Every 6 months for 2–3 years , then annually
NCCN	Every 3–6 months for 2 years , then annually

- Most guidelines are **risk-adapted; chest imaging (CT preferred) at the same intervals**.
- **Bloods:** renal function every visit; CBC and CMP if chemotherapy was given; **vitamin B12 annually from year 2**. Urine cytology every 6–12 months (low yield); **no biomarker is validated post-cystectomy**.
- **Urethral wash cytology is controversial:** asymptomatic detection shows a retrospective survival advantage; others reserve it for **multifocal tumours, CIS or prostatic urethral tumour**. If performed: **every 6–12 months in years 1–2**. Ask about urethral symptoms at every visit; **bloody urethral discharge → urethral wash or flexible cystoscopy**.
- **Diversion surveillance:** basic metabolic panel for **hypokalaemic hyperchloraemic metabolic acidosis** and renal dysfunction; **vitamin B12 deficiency** after bowel diversion; **stomal stenosis and parastomal hernia**; neobladder **incontinence or hypercontinence**; **urolithiasis**.

Follow-Up After Bladder Preservation

- **More intensive than after RC:** cystoscopy every 3–4 months in years 1–2, every 6 months in years 3–4, then annually to 10 years; CT urogram (or MRU) and chest CT every 3–6 months in years 1–2, then

annually in years 3–5; urine cytology every 6–12 months; bloods as after RC. Biopsy any local or distant recurrence.

Concept — *Surveillance intensity follows what urothelium remains: after cystectomy the threats are metastasis, the upper tract and the urethra; after bladder preservation the bladder itself must be watched for a decade.*

MANAGEMENT OF METASTATIC BLADDER CANCER

- Immune checkpoint inhibitors and targeted agents have transformed the field, but **cisplatin-based combination chemotherapy remains first-line** for metastatic UC; many patients face barriers to its delivery.
- **Cisplatin ineligibility (consensus working group — any one):** GFR <60 mL/min, grade ≥2 hearing loss, grade ≥2 neuropathy, WHO or ECOG performance status >1, NYHA class ≥III → **nearly 50% of patients** (Table 131.5).
- **Prognostic factors (Bajorin model, MVAC-treated series):** Karnofsky performance status <80% and visceral metastasis (lung, liver, bone) — median survival about 33, 13 and 9 months with 0, 1 and 2 factors. **Node-only disease: far better 5-year survival than visceral disease** (about 21% vs 7%).

First-Line Chemotherapy for Cisplatin-Eligible Patients

- **Regimens: ddMVAC and GC.**
- **Most (40–70%) respond initially, but most progress:** median survival about **14 months**, 5-year survival 5–20%.
- **Split-dose cisplatin** is occasionally used for **GFR 40–60 mL/min** when renal function is the only barrier — no randomised comparison with standard dosing.

TABLE 131.9 RANDOMIZED TRIALS OF FRONT-LINE SYSTEMIC THERAPY FOR PLATINUM-ELIGIBLE PATIENTS WITH METASTATIC UROTHELIAL CANCER

Trial	Comparison	Median survival (months)	Verdict
Intergroup	MVAC vs cisplatin	12.5 vs 8.2 (P = .0001); response 39% vs 12%	Combination beats single agent; more leukopenia, mucositis, neutropenic fever, drug-related death 3–4%
MD Anderson	MVAC vs CISCA	11.1 vs 8.3 (P = .0003)	MVAC better
Greek trial	Docetaxel-cisplatin vs MVAC	9.3 vs 14.2 (P = .026)	MVAC better
Carboplatin-paclitaxel trial	Carboplatin-paclitaxel vs MVAC	13.8 vs 15.4 (not significant)	No difference
von der Maase	GC vs MVAC	14.0 vs 15.2 (not significant); OS HR 1.04; response 49% vs 46%	Equivalent efficacy; GC better tolerated — fewer dose adjustments (37% vs 63%), toxic death 1% vs 3% ; MVAC: neutropenia, neutropenic sepsis, mucositis, alopecia; GC: grade 3–4 anaemia
EORTC 30924	High-dose (dose-dense) MVAC + granulocyte colony-stimulating factor (G-CSF), 2-weekly vs MVAC 4-weekly	15.5 vs 14.1 (P = .042); complete response 21% vs 9%; 7-year survival 24.6% vs 13.2%; HR 0.76	Dose-dense beats standard MVAC, but harder to tolerate

Trial	Comparison	Median survival (months)	Verdict
EORTC paclitaxel trial	Paclitaxel + GC vs GC	15.8 vs 12.7 (P = .075); response 55.5% vs 43.6%	No OS gain ; more neutropenia and thrombocytopenia
EORTC 30986 (cisplatin-ineligible)	Gemcitabine-carboplatin vs methotrexate, carboplatin, vinblastine (MCAVI)	9.3 vs 8.1 (not significant)	See cisplatin-ineligible below
Bevacizumab trial	GC ± bevacizumab	14.3 vs 14.5 (not significant)	No OS or PFS gain
Durvalumab ± tremelimumab phase III	Durvalumab + tremelimumab vs durvalumab vs chemotherapy	15.1 vs 14.4 vs 12.1 (not significant)	No OS gain , including PD-L1-positive
IMvigor130	Atezolizumab + gemcitabine-platinum vs chemotherapy (also atezolizumab alone)	16.0 vs 13.4 (P = .027) ; progression-free survival 8.2 vs 6.3 months	Slight benefit reported
KEYNOTE-361	Pembrolizumab + chemotherapy vs chemotherapy (also pembrolizumab alone)	17.0 vs 14.3 (P = .04) ; progression-free survival 8.3 vs 7.1 months	Not significant — the P-values did not meet the trial's prespecified thresholds → not recommended

- **Pattern:** only two chemotherapy advances have held — **combination over single agent**, and **dose-dense over standard MVAC**; GC matches MVAC with less toxicity; adding a third agent, bevacizumab or (as of this chapter) immunotherapy to chemotherapy has not reliably helped.
- **Trap:** GC is **not more effective** than MVAC — it is **equally effective and better tolerated**, which is why it is used more often.
- **Guidelines do not recommend first-line immunotherapy-chemotherapy combinations** on the basis of these three trials.

First-Line Chemotherapy for Cisplatin-Ineligible Patients

- **Carboplatin when cisplatin is contraindicated:** better tolerated but **less effective**; **never first-line for cisplatin-eligible patients**.
- **EORTC 30986: gemcitabine-carboplatin vs MCAVI** → equal median OS, **higher response (42% vs 30%)** and **less severe acute toxicity** → **gemcitabine-carboplatin is standard** in this group.
- Under study: gemcitabine-carboplatin + nivolumab vs gemcitabine-oxaliplatin + nivolumab (phase II); gemcitabine-paclitaxel (non-randomised).
- **Use cisplatin or carboplatin whenever the patient is platinum-eligible.**
- **Platinum-ineligible: performance status ≥2, GFR <30 mL/min, or multiple high-grade comorbidities** (Table 131.5 gives ≥3 for performance status) → poor outcomes regardless; immune monotherapy has been evaluated.

First-Line Immune Checkpoint Inhibitor Therapy

- Studied alone and with chemotherapy; checkpoint inhibitors already have neoadjuvant and adjuvant roles and have **improved survival** in advanced disease.

Immunotherapy for Platinum-Ineligible Patients

FIRST-LINE SINGLE-AGENT IMMUNOTHERAPY TRIALS

Trial	Agent	Population	Response
KEYNOTE-052 (phase II, single arm)	Pembrolizumab	Cisplatin-ineligible	Objective response 29%, complete 7%; higher with PD-L1 $\geq 10\%$
IMvigor210 (phase II, single arm)	Atezolizumab	Cisplatin-ineligible	Objective response 23%, complete 9%

- **Approval:** both approved first-line for **cisplatin-ineligible** patients in the United States and Europe; **only the US Food and Drug Administration (FDA) approves them for platinum-ineligible patients**. NCCN: **pembrolizumab** for platinum-ineligible; **atezolizumab** for platinum-ineligible with **PD-L1 expression**.
- **Trap:** first-line immunotherapy alone is **only for platinum-ineligible patients**; anyone eligible for cisplatin or carboplatin receives **platinum first**.

Maintenance Immunotherapy After Platinum-Based Chemotherapy

- **JAVELIN Bladder 100** (phase III): patients **without progression after first-line gemcitabine-platinum** randomised to **avelumab maintenance vs no further treatment** → **median OS 21.4 vs 14.3 months (HR 0.69)**; OS rate 71.3% vs 58.4%; more immune-related adverse events.
- Pembrolizumab maintenance (phase II): OS end point not met.
- **Standard of care: avelumab maintenance for stable or responding disease after first-line platinum chemotherapy.**

Immunotherapy-Chemotherapy Combination Approaches

- Covered in Table 131.9: **IMvigor130** (atezolizumab — a PD-L1 inhibitor — with gemcitabine-platinum) gave a slight benefit; **KEYNOTE-361** (pembrolizumab) and the **durvalumab ± tremelimumab** trial did not → **combinations not recommended first-line**.

Second-Line Therapy

- **No clear standard second-line chemotherapy**; patients failing first-line chemotherapy have a poor prognosis, and conventional salvage agents give suboptimal responses — **immunotherapy has changed this**.

Second-Line Chemotherapy

- **Re-treat with platinum combination (GC or ddMVAC) if the last platinum was ≥ 12 months earlier**; response is better in prior responders (MVAC after GC: about 44% vs 14%), and platinum-combination re-treatment gives better OS than non-platinum chemotherapy.
- **Single agents (NCCN alternatives): paclitaxel, docetaxel, gemcitabine** — modest responses and survival (Table 131.10); **docetaxel causes myelosuppression requiring dose reduction in most**.
- **Gemcitabine-paclitaxel:** responses vary widely (0–60%) by schedule; a phase III trial of short vs prolonged therapy found **no OS or progression-free survival difference**.
- **Vinflunine** (antitubulin derived from a vinca alkaloid) vs best supportive care → low response (8.6%) but **OS benefit on updated analysis (HR 0.72)** → **approved in Europe (2009) as second-line; not used in the United States**.
- **Larotaxel** (semisynthetic taxane, claimed activity in taxane-resistant cells and blood–brain barrier penetration) + cisplatin vs GC first-line: closed early (sponsor decision); **no OS difference, worse progression-free survival** → no gain over GC.

Second-Line Immune Checkpoint Inhibitor Therapy

- **Pembrolizumab is the preferred second-line agent** after progression on, or refractoriness to, platinum; alternatives: nivolumab, avelumab.
- **KEYNOTE-045 (phase III):** pembrolizumab vs investigator's choice (paclitaxel, docetaxel or vinflunine) → median OS 10.3 vs 7.4 months, irrespective of PD-L1, with fewer treatment-related adverse events → FDA approval; level 1 evidence.
- **Atezolizumab:** favourable phase II (IMvigor210, post-platinum cohort) → **phase III IMvigor211: no OS benefit even with high PD-L1** → **not recommended** in this setting.
- **Nivolumab (CheckMate 275), avelumab (JAVELIN Solid Tumor) and durvalumab:** responses in about a fifth, independent of PD-L1; avelumab durable and well tolerated (infusion reactions, fatigue).

TABLE 131.10 TRIALS OF SECOND- AND THIRD-LINE THERAPY FOR METASTATIC UROTHELIAL CANCER

Agent (trial)	Eligibility	Response rate	Median OS (months)
Paclitaxel, weekly	One prior regimen	10%	7.2
Docetaxel, 3-weekly	One prior cisplatin regimen	13%	9.0
Gemcitabine	One prior platinum or cisplatin regimen	11–22.5%	5.0–8.7
Gemcitabine-paclitaxel	One prior cisplatin regimen	41.5–60%	14.4
Vinflunine vs best supportive care	Prior platinum	8.6%	6.9 vs 4.6
Pembrolizumab vs chemotherapy (KEYNOTE-045)	Prior platinum	21.1% vs 11.4%	10.3 vs 7.4
Nivolumab (CheckMate 275)	Prior platinum	19.6%	8.7
Durvalumab	Prior platinum	17.8%	Not given
Atezolizumab vs chemotherapy (IMvigor211)	Prior platinum	Not given	11.1 vs 10.6 (not significant)
Avelumab (JAVELIN)	Prior platinum	17%	Not given
Enfortumab vedotin vs chemotherapy (phase III)	Prior platinum and immunotherapy	40.6% vs 17.9%	12.88 vs 8.97
Sacituzumab govitecan	Prior platinum and immunotherapy	27%	10.9
Erdafitinib	FGFR-altered; prior platinum and/or immunotherapy	40% overall (body text); 59% (table — the post-immunotherapy subgroup)	11.3

- **What the table tests:** only pembrolizumab and enfortumab vedotin show phase III OS wins over chemotherapy after platinum; atezolizumab's phase II response **did not translate** into survival.
- **Trap:** response rate is not survival.

Immune Checkpoint Inhibitor Side Effects

- **Mechanism:** immune over-activation. Commonest: rash, colitis/diarrhoea, hepatitis, pneumonitis, and thyroid, adrenal, pancreatic or pituitary dysfunction; rarely myocarditis.

- **Endocrine toxicities are often irreversible** → counsel and manage promptly (American Society of Clinical Oncology guidance).

Antibody Drug Conjugates

- **Antibody–drug conjugates (ADCs)** = monoclonal antibody linked to a cytotoxic agent → bind tumour-selective antigens → deliver chemotherapy selectively → **greater efficacy, less toxicity**.

ANTIBODY–DRUG CONJUGATES AND FGFR INHIBITION

Agent	Target	Payload / class	Signature toxicities	Position
Enfortumab vedotin (EV)	Nectin-4 (cell adhesion molecule)	Monomethyl auristatin E	Alopecia, fatigue, peripheral neuropathy , rash, anorexia, nausea, diarrhoea	After platinum and immunotherapy; preferred after immunotherapy in platinum-ineligible; with pembrolizumab: preferred first-line for cisplatin-ineligible (NCCN)
Sacituzumab govitecan	Trophoblast cell-surface antigen 2 (Trop-2)	SN-38	Neutropenia , fatigue, alopecia, gastrointestinal	Third line — accelerated FDA approval after platinum and immunotherapy
Erdafitinib	FGFR (oral tyrosine kinase inhibitor)	—	Hyperphosphataemia , stomatitis, asthenia	Second line after platinum or immunotherapy for FGFR3 or FGFR2 alterations

- **EV phase II (post-platinum and immunotherapy)**: objective response 44%, complete 12%, median duration 7.6 months; cisplatin-ineligible cohort after immunotherapy only: response 52%, complete 20%.
- **EV phase III vs second-line chemotherapy**: survival benefit (Table 131.10) → **approved in the United States and Europe** after platinum and checkpoint inhibitor, or for cisplatin-ineligible patients after one or more prior lines.
- **EV + pembrolizumab, first line, cisplatin-ineligible (phase Ib/II)**: objective response 73%, complete 16%; median response duration 25.6 months; OS 26.1 months → phase III under way.
- **Sacituzumab govitecan**: objective response 27%, with most patients showing a reduction in tumour burden; median OS 10.9 months.

Fibroblast Growth Factor Receptor Inhibition

- **Erdafitinib** (phase II, **FGFR3 mutation or FGFR2/3 fusion**, post-platinum): objective response 40% plus stable disease 39%; 59% in those with prior immunotherapy; median OS 11.3 months. Further FGFR inhibitors are in trials.

Vascular Endothelial Growth Factor Inhibitor Therapy

- **Vascular endothelial growth factor (VEGF) receptors are expressed in UC**; VEGF recruits circulating endothelial progenitors → **angiogenesis**, essential for growth and metastasis; urinary and serum VEGF correlate with metastatic potential and recurrence.

TARGETED AGENTS WITHOUT POSITIVE OUTCOMES

Target	Agent	Outcome
VEGF	Bevacizumab (humanised monoclonal antibody)	Phase II with GC promising (response 72%) → phase III: no OS or PFS gain → not recommended

Target	Agent	Outcome
VEGF receptors (multikinase)	Sunitinib (oral tyrosine kinase inhibitor)	With GC: unacceptable toxicity ; maintenance phase II: no PFS gain (closed early, underpowered)
Epidermal growth factor receptor (EGFR) — overexpressed, correlates with stage and grade	Cetuximab, gefitinib, lapatinib	Historically used; not recommended
Human epidermal growth factor receptor 2 (HER2)	Trastuzumab	High response with paclitaxel-gemcitabine-carboplatin in HER2-positive tumours, but randomised phase II: no benefit added to gemcitabine-platinum

FIRST-LINE TO THIRD-LINE SEQUENCING IN METASTATIC UROTHELIAL CARCINOMA

Line	Cisplatin-eligible	Cisplatin-ineligible, platinum-eligible	Platinum-ineligible
First	GC or ddMVAC	Gemcitabine-carboplatin, or EV + pembrolizumab (NCCN preferred)	Pembrolizumab ; atezolizumab if PD-L1-positive
Maintenance	Avelumab if no progression after platinum	Avelumab if no progression after platinum	—
Second	Pembrolizumab (preferred); nivolumab, avelumab; erdafitinib if FGFR2/3-altered; platinum re-challenge if ≥ 12 months	As for cisplatin-eligible; EV	EV (preferred after immunotherapy)
Third	EV; sacituzumab govitecan	EV; sacituzumab govitecan	—

Future Directions

- **Biomarker-driven risk stratification** (tumour, urine, serum) and **combination therapies** across neoadjuvant, adjuvant and advanced settings are expected to keep improving survival.

Concept — Metastatic treatment is **sequenced by platinum eligibility**: platinum first for anyone who can take it, then avelumab to hold the response, then checkpoint inhibition, antibody–drug conjugates or FGFR inhibition at progression — immunotherapy alone first-line is the exception reserved for the platinum-ineligible.

KEY POINTS

- **Optimal MIBC care is multidisciplinary** (urological, medical and radiation oncology); clinical staging lacks the sensitivity and specificity to predict final pathological and nodal stage — a major limit on pretreatment decisions.
- **Gold standard for organ-confined MIBC: radical cystectomy**, with **neoadjuvant cisplatin-based chemotherapy** for eligible patients.
- **Multimodal bladder preservation** (chemoradiation with maximal TUR) gives outcomes comparable to cystectomy in **carefully selected patients** (clinical T2, small solitary tumour, no hydronephrosis, no concurrent CIS).
- Despite aggressive surgery, **about 50% of cystectomy patients ultimately die of disease**.
- **Neoadjuvant cisplatin-based chemotherapy** gives a modest but significant survival advantage on meta-analysis; adjuvant evidence is weaker, so **neoadjuvant chemotherapy should be considered in all cisplatin-eligible patients**.

- **Adjuvant nivolumab** benefits patients with high disease burden after cystectomy who received neoadjuvant cisplatin or are cisplatin-ineligible; other adjuvant immunotherapies have not matched it.
- In metastatic disease, **initial cisplatin responses are favourable but rarely durable**; outcomes are poor.
- **Improved or stable metastatic disease after platinum → maintenance avelumab.**
- **Progression on platinum → pembrolizumab second line.**
- **Antibody-drug conjugates (enfortumab vedotin) and other targeted therapies** are FDA-approved options after platinum and/or immunotherapy.